FBISE Math Class 9th Solved New Model Paper 2026 Federal Board New Pattern 2026 50% MCQs & 50% Subjective SEC B & C

1) Using the laws of exponents, simplify:

$$\left(\frac{2187\times a^5\times b^{17}}{a^{12}\times b^3}\right)^{\frac{1}{7}}$$

Sol:

$$= \left(\frac{3^7\times a^5\times b^{17}}{a^{12}\times b^3}\right)^{\frac{1}{7}}$$

$$= \left(\frac{3^7\times b^{17}\times b^{-3}}{a^{12}\times a^{-5}}\right)^{\frac{1}{7}}$$

$$= \left(\frac{3^7\times b^{14}}{a^7}\right)^{\frac{1}{7}}$$

$$= \frac{(3^7)^{\frac{1}{7}}\times(b^{14})^{\frac{1}{7}}}{(a^7)^{\frac{1}{7}}}$$

$\frac{1}{\cancel{7}}\times \cancel{14}^{2}$

$$= \frac{3\times b^2}{\quad}$$

$$\boxed{\left(\frac{2187\, a^5 b^{17}}{a^{12} b^3}\right) = \frac{3b^2}{a}}$$

End

Verify

$a = 2,\ b = 3 \Rightarrow \frac{3\times 9}{2}$

$\frac{27}{2}$

(OR)

(1) OR

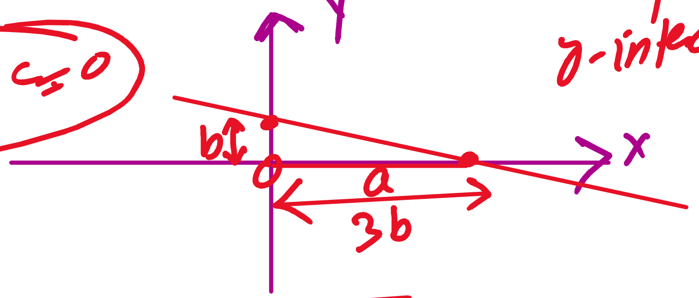
A line cuts intercepts on axes such that x-intercept is three times y-intercept and the line pass through $P(3,2)$. Write the equation of the line in standard form.

$a = 3b$

$x \to a$ intercep

y-intercept $\to b$

$ax + by + c = 0$



Sol: As Eqn. of straight line in two-intercepts form is

$$\frac{x}{a} + \frac{y}{b} = 1 \to (1)$$

$a = x$-inter.

$b = y$-inter.

a = x-inter.
b = y-inter.

$a \qquad b$

According to given condition

$$a = 3b \rightarrow \text{given}$$

Eqn. (1) $\Rightarrow$

$$\frac{x}{3b} + \frac{y}{b} = 1 \rightarrow (2)$$

As this line (2) passes through $P(3, 2)$ [x, y] so, it will satisfy eqn

$$\Rightarrow \frac{\cancel{3}}{\cancel{3}b} + \frac{2}{b} = 1$$

$$\Rightarrow \frac{1}{b} + \frac{2}{b} = 1$$

$$\Rightarrow \frac{1+2}{b} = 1$$

$$\frac{3}{b} = 1$$

$$\Rightarrow \boxed{b = 3} \rightarrow \text{y-intercept}$$

Put value of b in Eqn. (2)

$$\frac{x}{9} + \frac{y}{3} = 1$$

x by 9 on B.S.

$$\boxed{x + 3y = 9}$$

or

$$\boxed{x + 3y - 9 = 0}$$

is the req. Eqn. of straight line in standard form

2	Solve the equation $\mid 2x - 3 \mid = \mid x + 5 \mid$

$$|x| = \begin{cases} +x, & x > 0 \\ -x & x < 0 \end{cases}$$

$$|-5| = -x = -(-5) = 5$$

Sol:

$$2x - 3 = \pm(x + 5)$$

$2x - 3 = x + 5$

$2x - x = 3 + 5$

$$\boxed{x = 8}$$

OR

$2x - 3 = -(x + 5)$

$2x - 3 = -x - 5$

$2x + x = 3 - 5$

$3x = -2$

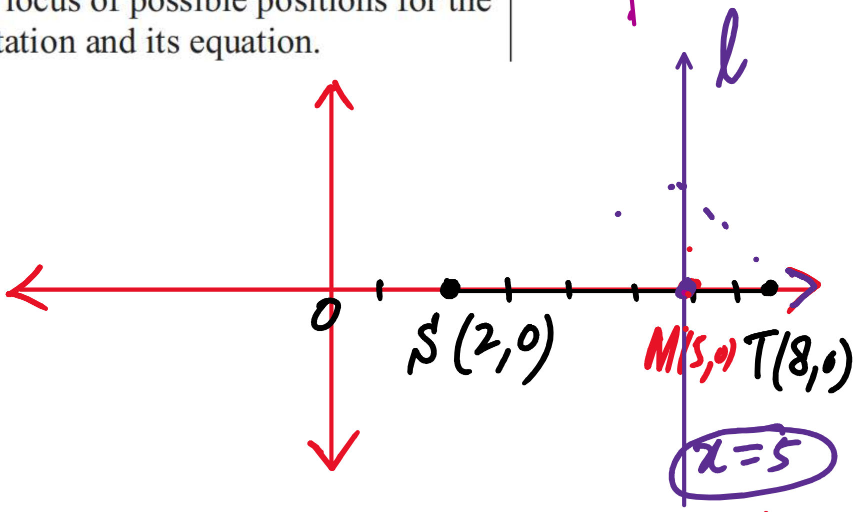
$3x = -2$

$\Rightarrow \boxed{x = -\frac{2}{3}}$

$S.S = \{8, -\frac{2}{3}\}$ A2

OR	Two power stations are at $S(2,0)$ and $T(8,0)$. A substation is to be built equidistant from S and T. Find locus of possible positions for the substation and its equation.	3

Sol:-



The substation will be built on the perpendicular bisector of $\overline{ST}$

Mid-point of $\overline{ST} = M(x, 0)$

$x = \frac{2+8}{2} = \frac{10}{2} = 5$

$y = \frac{0+0}{2} = 0$

So $M(5,0)$ is the mid-point

So $M(5,0)$ is the midpoint

of $\overline{ST}$

Possible positions

The substation can be placed anywhere on the bisector of $\overline{ST}$

hence there are infinite positions for the substation

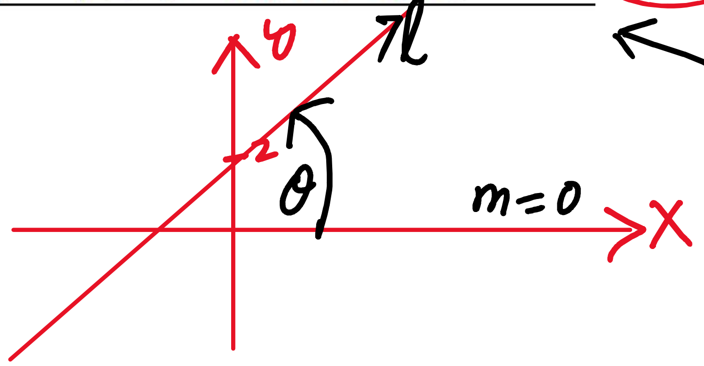
Eqn. of the line $\Rightarrow \boxed{x=5}$

$y = mx + c$

3	Find the acute angle between the line $y = \sqrt{3}x + 2$ and the x-axis.

$< 90^\circ$

Sol:



$m =$ slope $=$ gradient $= \tan\theta$

As $y = \sqrt{3}\,x + 2$

$m = \sqrt{3}$

$$\boxed{m = \sqrt{3}}$$

$$\Rightarrow \tan\theta = \sqrt{3}$$

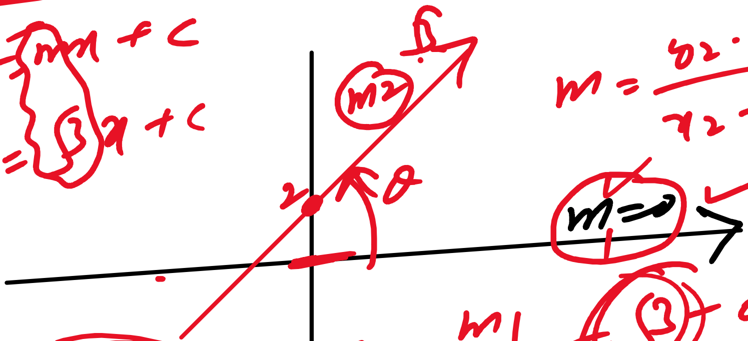
$$\theta = \tan^{-1}(\sqrt{3})$$

$$\boxed{\theta = 60^\circ}$$

$y = mx + c$

$y = \sqrt{3}x + c$

$(3, -3)$



$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\tan\theta = \frac{m_2 - m_1}{1 + m_1 m_2} = \frac{\sqrt{3} - 0}{1 + 0}$$

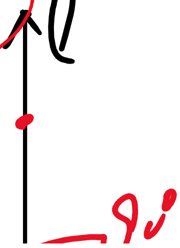
MQ

Check point

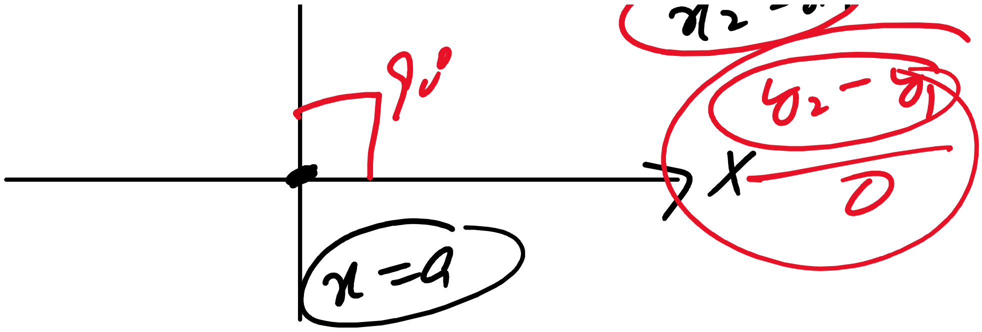
The slope of vertical line is

a) 0 b) 1 c) -1 d) ∞

undefined

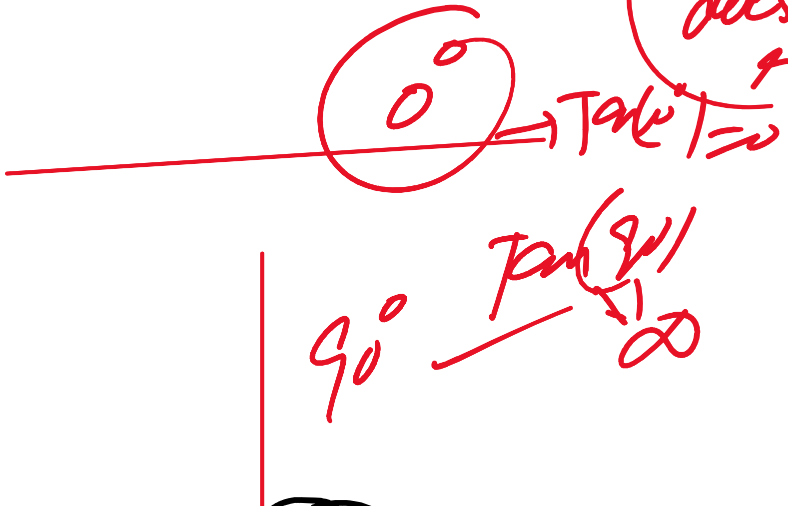


$$m = \frac{y_2 - y_1}{x_2 - x_1}$$



Horizontal line slope = 0

vertical " " = ∞ or undefined, does not exist



Prove that $\sqrt{\frac{1-cos\theta}{1+cos\theta}} = \frac{sin\theta}{1+cos\theta}$

Sol:- $L.H.S = \sqrt{\frac{1-\cos\theta}{1+\cos\theta}}$

$$= \sqrt{\frac{1-\cos\theta}{1+\cos\theta} \times \frac{1+\cos\theta}{1+\cos\theta}}$$

$$= \sqrt{\frac{1-\cos^2\theta}{(1+\cos\theta)^2}}$$

$\sqrt{(1+\cos\alpha)^2}$

$= \dfrac{\sqrt{\sin^2\theta}}{\sqrt{(1+\cos\alpha)^2}} = \dfrac{\sin\theta}{1+\cos\alpha}$

$= \underline{R.H.S}$

$\dfrac{x^3}{x}$ Hence $L.H.S = R.H.S$

$= x^2+5x+6$

$\dfrac{(x+3)(x+1)(x+2)}{(x+1)}$

4	The volume of a rectangular box is given by $x^3+6x^2+11x+6$ cubic cm. If its height is $(x+1)$ cm, find the product of its length and width.

As Volume = $\boxed{\text{length} \times \text{width}}$ $\times$ height

$x^3+6x^2+11x+6 = \text{length} \times \text{width} \times (x+1)$

$\Rightarrow \text{length} \times \text{width} = \dfrac{x^3+6x^2+11x+6}{x+1}$

$\dfrac{5x^2}{x}$

x^2+5x+6 ✓

$x+1 \overline{)\, x^3+6x^2+11x+6}$

$\quad \ominus x^3 \ominus x^2$

$\quad 5x^2+11x+6$

$\quad \ominus 5x^2 \ominus 5x$

$5x^2 + 5x$
$- \quad -$

$6x + 6$
$6x + 6$
$- \quad -$

0

Hence $\boxed{\text{length} \times \text{width} = x^2+5x+6}$

Pr

Solve the inequality $-5 < \frac{2x+7}{3} \leq 1$ and represent the solution set on the real line.

$x = ?$

Sol:

$$-5 < \frac{2x+7}{3} \leq 1$$

$\times 3$ on both sides, we get

$$-15 < 2x+7 \leq 3$$

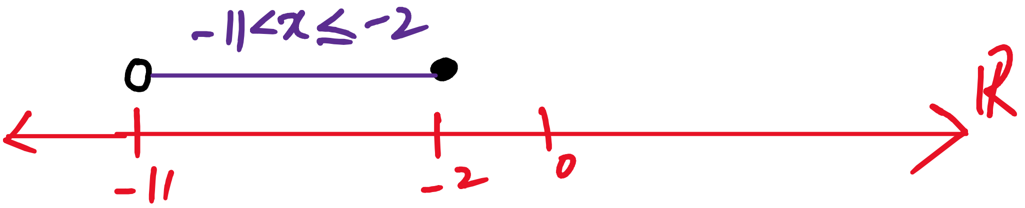
Adding -7, we get

$$\Rightarrow -15-7 < 2x \leq 3-7$$

$$-22 < 2x \leq -4$$

$\div$ by 2, we get

$$\boxed{-11 < x \leq -2}$$



Alternate method

Sol: As $-5 < \dfrac{2x+7}{3} \leq 1$ is composite inequality $\Rightarrow$

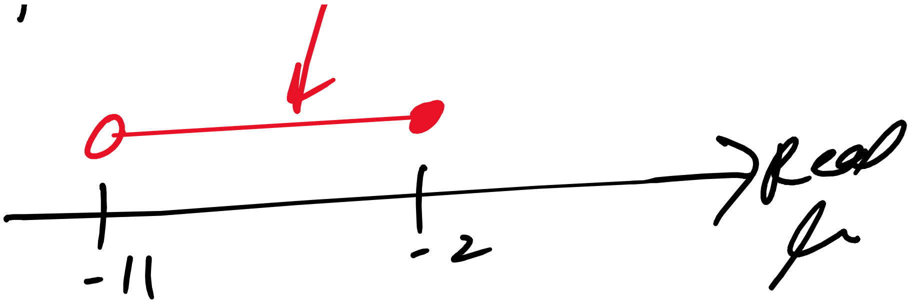
$\dfrac{2x+7}{3} > -5$ And $\dfrac{2x+7}{3} \leq 1$

$2x+7 > -15$ and $2x+7 \leq 3$

$2x > -22$ $\quad$ $2x \leq -4$

$\boxed{x > -11}$ and $x \leq -2$

$$\boxed{-11 < x \leq -2}$$



5	In a survey, 200 people were asked if they prefer tea. 85 said yes. Calculate the relative frequency of preferring tea.

Sol: Relative freq of preferring

Tea $= RF(\text{Tea}) = \dfrac{85}{200}$

$= \dfrac{17}{40} = 0.425$

$\boxed{RF(\text{preferring Tea}) = 42.5\%}$

Check: If $P(E) = 0.35$, then this event is $\longrightarrow$?

a) Unlikely b) Likely c) Equally likely

d) Sure

	The intensity of sound is measured by $L = 10\log\left(\frac{I}{I}\right)$.

I = Intensity

OR	The intensity of sound is measured by $L = 10\log\left(\frac{I}{I_0}\right)$. A machine produces intensity $1000I_0$. Find the value of L.

$I =$ Intensity

$I = 1000\, I_0$

Sol:

$$L = 10\log\left(\frac{I}{I_0}\right)$$

$$\Rightarrow L = 10\log\left(\frac{1000\cancel{I_0}}{\cancel{I_0}}\right)$$

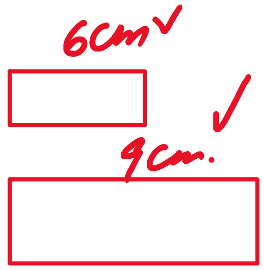
$$L = 10\log(1000)$$

$$L = 10\log(10^3)$$

$$L = 3\times 10\log(10)$$

$$\Rightarrow \boxed{L = 30}$$

6	Two similar rectangles have lengths 6cm and 9 cm. (a) Find ratio of the corresponding sides. (b) Find ratio of the corresponding areas.



Sol: (a) $l_1 : l_2 = 6 : 9$

$$\Rightarrow \frac{l_1}{l_2} = \frac{6}{9} = \frac{2}{3}$$

$$\Rightarrow \boxed{l_1 : l_2 = 2 : 3}$$

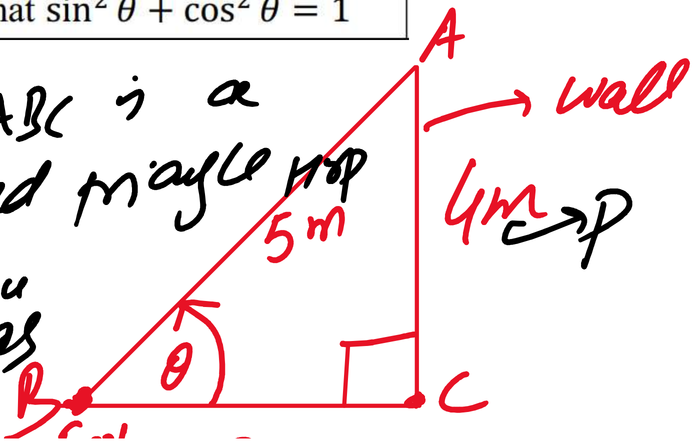
(b) $A_1 : A_2 = (l_1 : l_2)^2$

$$= (2:3)^2$$

$$\boxed{A_1 : A_2 = 4 : 9}$$

IA $V_1 : V_2 = (2:3)^3 = 8 : 27$

OR	A ladder reaches a height of 4 m on a wall. The foot of the ladder is 3 m from the wall. If θ is the angle the ladder makes with the ground, (a) Find the values of $\sin\theta$ and $\cos\theta$. (b) Verify that $\sin^2\theta + \cos^2\theta = 1$



Sol.. As ΔABC is a right-angled triangle

By Pythagoras

By Pythagoras Theorem

B (foot) — θ — C, 3m

a) $AB^2 = BC^2 + AC^2$

$\sqrt{AB^2} = \sqrt{3^2 + 4^2} = \sqrt{9+16}$

$\Rightarrow \boxed{AB = 5\,m}$

$\boxed{\sin\theta = \frac{4}{5}}$ $\boxed{\cos\theta = \frac{3}{5}}$

b) $L.H.S = \sin^2\theta + \cos^2\theta$

$= \left(\frac{4}{5}\right)^2 + \left(\frac{3}{5}\right)^2$

$= \frac{16}{25} + \frac{9}{25}$

$= \frac{16+9}{25} = \frac{25}{25} = 1$

$= R.H.S$

$\Rightarrow L.H.S = R.H.S$

$y = f(x)$

P(x, y)

D, R

7	Consider the relation $R = \{(2,4), (3,9), (4,16), (5,25)\}$. (a) Find the domain and range of the	3

7	Consider the relation $R = \{(2,4),(3,9),(4,16),(5,25)\}$. (a) Find the domain and range of the relation. (b) Identify the rule connecting the elements.	3

(D, R)

$2 \times 2 = 2^2$

$3 \times 3 = 9$

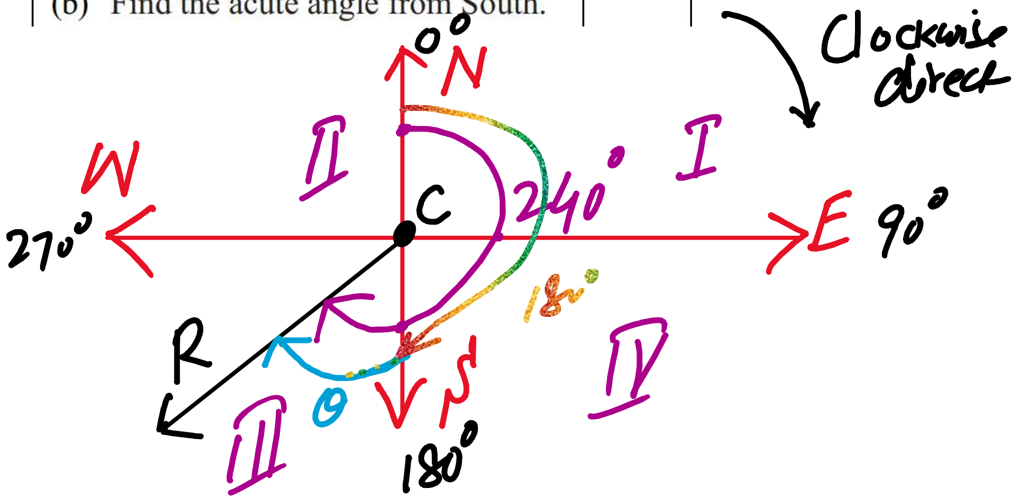
x

Sol. Dom $R = \{2, 3, 4, 5\} \to x$

a) Range $R = \{4, 9, 16, 25\} \to y$

b) Rule: $\boxed{y = x^2}$

OR	A scout walks from camp C to a river R on a bearing of 240°. Draw a rough sketch showing the direction. (a) State in which quadrant the bearing lies? (b) Find the acute angle from South.	3



a) 3rd quadrant as 240° lies s/w South and West

b) Acute angle from South

b) Acute angle from south to CR is $= 240^\circ - 180^\circ = 60^\circ$

$A = \{1, 2, 3\}$ $B = \{3, 4, 5\}$

$A \cup B = \{1, 2, 3, 4, 5\}$

$A \cap B$

[A, B] → $A \cup B$

SECTION – C (Marks 18)

(3 × 6 = 18)

Note: Attempt all questions. Marks of each question are given.

Q. No.	Question	Marks		Question	Marks
8	Verify the Associative Laws of union for three sets A, B, C using the analytical method.	6	*OR*	Two similar metal spheres have radii in the ratio 5 : 2. Find the ratio of their surface areas and volumes. If the smaller sphere has volume 40πcm³, find the larger sphere's volume.	6

Q. No.	Question
8	Verify the Associative Laws of union for three sets A, B, C using the analytical method. $\in$

We have to prove

$A \cup (B \cup C) = (A \cup B) \cup C$

Proof:- ~~L.H.S = A ∪ (B ∪ C)~~

Let $x \in A \cup (B \cup C)$, then

$x \in A$ or $x \in (B \cup C)$

$\Rightarrow x \in A$ or $(x \in B$ or $x \in C)$

$\Rightarrow (x \in A$ or $x \in B)$ or $x \in C$

$x \in (A \cup B)$ or $x \in C$

$\Rightarrow x \in (A \cup B) \cup C$

Thus $x \in A \cup (B \cup C) = x \in (A \cup B) \cup C$

$\Rightarrow A \cup (B \cup C) = (A \cup B) \cup C$

OR	Two similar metal spheres have radii in the ratio $5:2$. Find the ratio of their surface areas and volumes. If the smaller sphere has volume $40\pi \text{cm}^3$, find the larger sphere's volume.

Sol: $r_1 : r_2 = 5:2$

Ratio of surface Areas

$A_1 : A_2 = (5:2)^2$

$\boxed{A_1 : A_2 = 25:4}$

$V_1 : V_2 = (5:2)^3$

$$V_1 : V_2 = (5:2)^3$$

$$\Rightarrow \boxed{V_1 : V_2 = 125 : 8} \rightarrow (1)$$

Given $V_2 = 40\pi \text{ cm}^3$

$V_1 = ?$

$$\Rightarrow \frac{V_1}{V_2} = \frac{125}{8}$$

$$\Rightarrow \frac{V_1}{40\pi} = \frac{125}{8}$$

$$\Rightarrow V_1 = \frac{125}{8} \times 40\pi$$

$$\boxed{V_1 = 625\pi \text{ cm}^3}$$

$x = \frac{1}{2} \Rightarrow 2x = 1$ $(2x-1)$ $x = -3$ $(x+3)$

4/3	9	If $A = 12x^3y^2(x^2-9)$, $B = 18x^2y^3(2x^2+5x-3)$ Factorize both expressions	(x^2-3^2) 2 2

9	If $A = 12x^3y^2(x^2-9)$, $B = 18x^2y^3(2x^2+5x-3)$ Factorize both expressions completely, then determine their HCF and LCM. Also verify that A × B = HCF × LCM	

$2\,|\,12$, $2\,|\,6$, $3\,|\,3$, 1

$3\,|\,9$, $3\,|\,3$, 1

Factorization

Sol:

$$A = 2^2 \cdot 3^1 \cdot x^3 \cdot y^2 (x-3)(x+3)$$

$$B = 2^1 \cdot 3^2 \cdot x^2 y^3 (x+3)(2x-1)$$

$2x^2+5x-3$

$(6)(-1)$

$(2)(-3) = -6$

6×1

2×3

$2x^2+6x-x-3$

$2x(x+3) - 1(x+3) = (x+3)(2x-1)$

$$HCF = 2 \cdot 3 \cdot x^2 \cdot y^2 (x+3)$$

$$\Rightarrow \boxed{HCF = 6x^2y^2(x+3)}$$

$$LCM = 2^2 \cdot 3^2 \cdot x^3 \cdot y^3 (x+3)(x-3)(2x-1)$$

$$\boxed{LCM = 36x^3y^3(x^2-9)(2x-1)}$$

$A \times B = 216x^5y^5(x^2-9)(2x^2+5x-3)$

$$\boxed{A \times B = 216x^5y^5(x^2-9)(2x^2+5x-3)}$$

$$\boxed{HCF \times LCM = 216x^5y^5(x^2-9)(x+3)(2x-1)}$$

$$\boxed{HCF \times LCM = 216x^5y^5(x^2-9)(2x^2+5x-3)}$$

Hence $A \times B = HCF \times LCM$

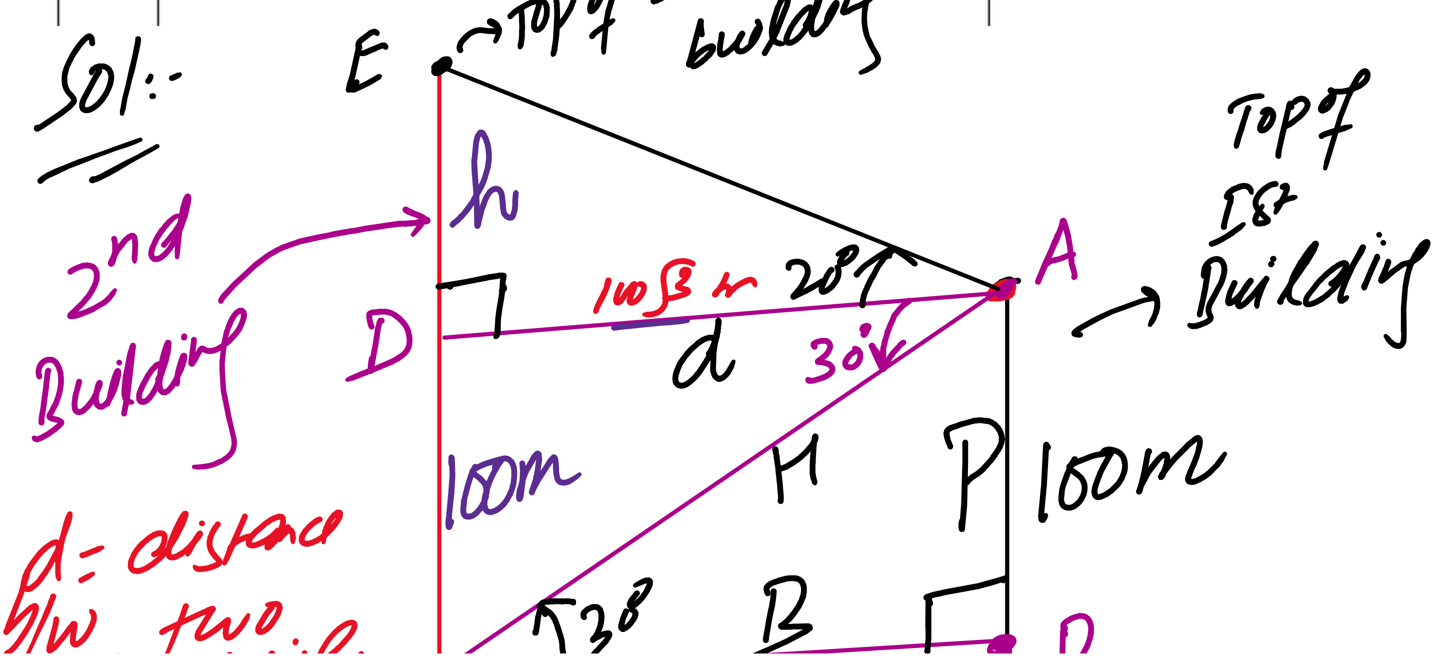
proved

OR	From the top of a 100 m high building, the angle of depression to the bottom of a second building is 30°and the angle of elevation to the top of the second building is 20°. Calculate the height of the second building.	6

↑ elevation

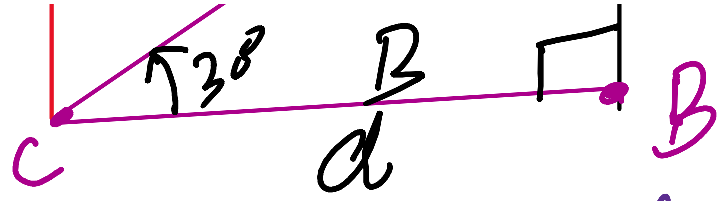
↙ dep

Sol:-



d = distance b/w two ...

b/w two buildings



Find CE → Total height of the 2nd building

$$H = CE = (h + 100)\,m \rightarrow (1)$$

In right-angled $\triangle ABC$,

$$\tan 30^\circ = \frac{100m}{d}$$

$$\Rightarrow d = \frac{100m}{\tan 30^\circ}$$

$$d = \frac{100m}{\frac{1}{\sqrt{3}}}$$

$$\boxed{d = 100\sqrt{3}\ m}$$

In Right-angled $\triangle ADE$;

$\tan 20^\circ = \frac{h}{d}$

$\Rightarrow h = \tan 20^\circ \times 100\sqrt{3}\,m$

$\boxed{h = 63.04\,m}$

Total Height $H = CE = (h + 100)\,m$

$\boxed{\text{Total Height} = CE = 163.04\,m}$

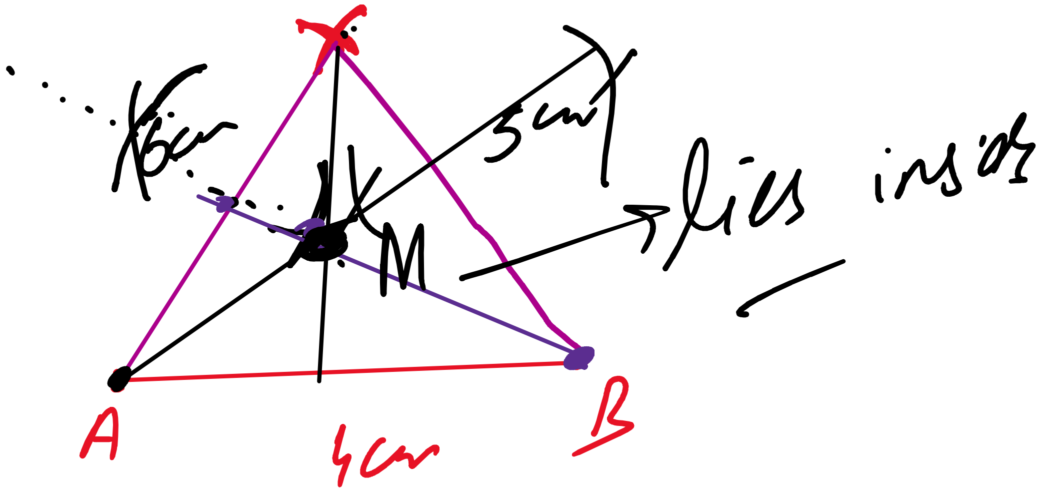
Ans

10	Draw a triangle ABC with $AB = 4$ cm, $BC = 5$ cm, and $CA = 6$ cm. Construct all three medians of the triangle. Verify that they are concurrent. Name the point of concurrency, and state whether it lies inside, on, or outside the triangle.	6

C

4cm

Sol:



OR	A school principal wants to allocate funds to different departments based on student enrolment. The number of students in each department is tabulated as under.	6

Subject	x	Science	Maths	English
No. of Students	f	200	150	180

If the total available fund is Rs. 700,000, calculate the amount each department should receive if funds are distributed equally per student.

Rs

$\frac{100}{5}$ Sol

Total Funds = 700,000

Total Students = 200 + 150 + 180
= 530

Allocation of funds per student = $\frac{700,000}{530}$

Science Funds = $200 \times \frac{700,000}{530}$ =

Science funds = 200 X $\frac{700,000}{530}$ = ✓

Math funds = 150 x $\frac{700,000}{530}$ = ✓

English funds = 180 x $\frac{700,000}{530}$ = ✓